

Multilingual AI assistant for public service access

Project description, workflow and 8-week plan

1. Project description

This project aims to develop an AI-powered multilingual intelligent assistant that helps citizens easily obtain information about government schemes, education, healthcare, agriculture, legal procedures and financial services through natural conversation. Users can interact using text or voice in their preferred language, making the system accessible to people with varying levels of digital literacy.

The assistant will combine Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Speech-to-Text (STT) and Text-to-Speech (TTS) to understand each question, retrieve information from trusted sources and give accurate, easy-to-understand answers.

In the initial phase the system will support a few widely used Indian languages and will be designed so that more languages and service domains can be added later. It will be delivered as a web-based platform with integrated chat and voice. By reducing language barriers and simplifying access to reliable information, the project promotes digital inclusion, improves awareness of public services and helps citizens make informed decisions about their rights, benefits and opportunities.

Objectives

- Let citizens ask questions by text or voice in their own language.
- Answer only from trusted, official sources and show where each answer came from.
- Give short, simple answers suitable for users with low digital literacy.
- Admit uncertainty and point to the official office or helpline instead of guessing.
- Keep the design scalable so new languages and domains can be added later.

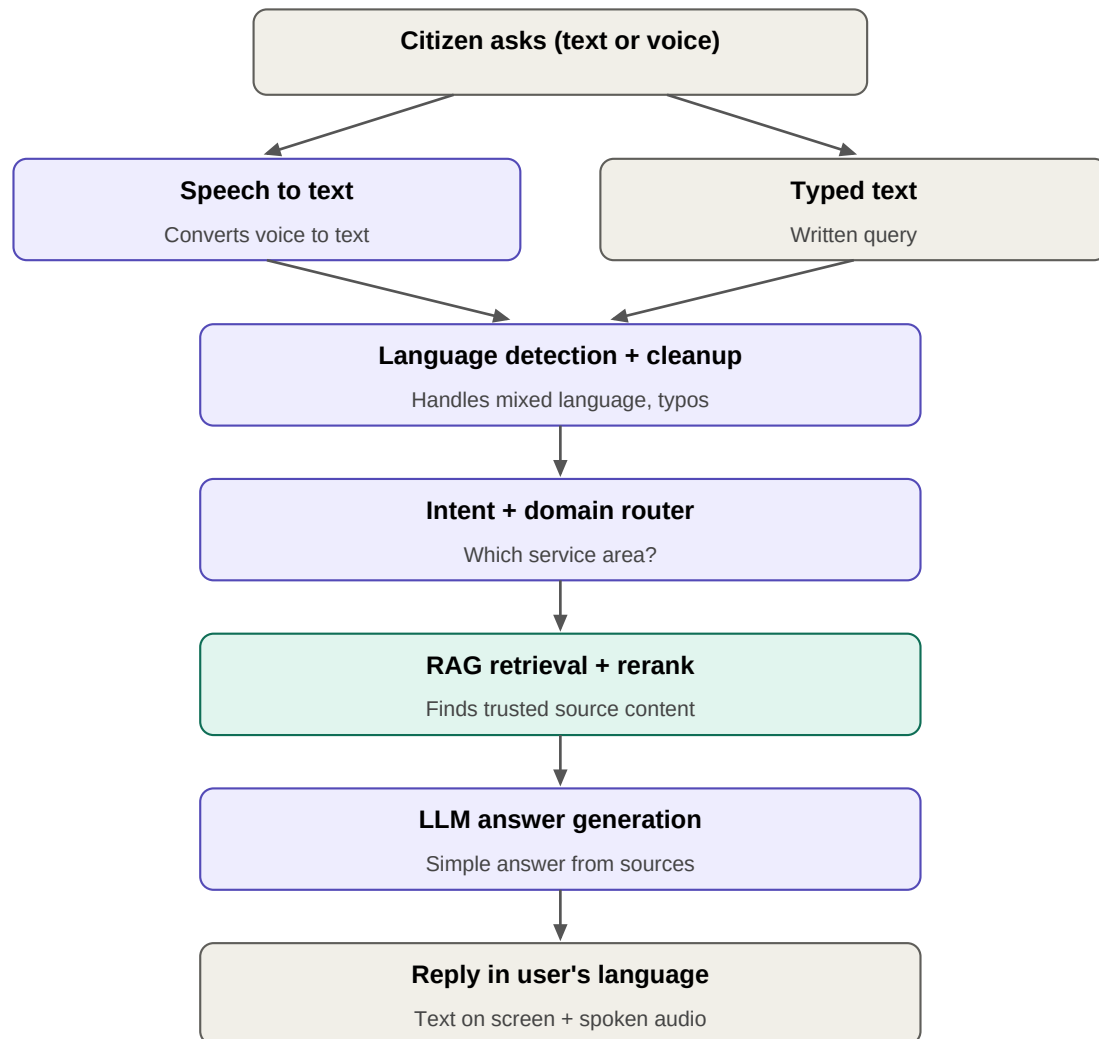
Scope of the first version

- A small number of languages (for example English, Hindi and one regional language).
- Two service domains (for example government schemes and agriculture).
- A limited, carefully chosen set of official documents as the knowledge base.
- A web interface with chat, voice input and voice output.
- Technologies will be finalised in week 1 after small trials; the plan below does not depend on any one tool.

2. AI workflow

2.1 Runtime flow: what happens for each question

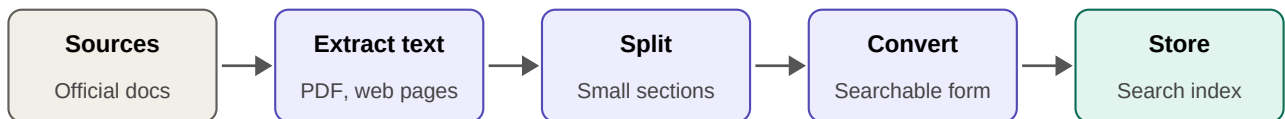
Colour key: grey = user-facing, purple = AI model step, teal = knowledge base.



How the workflow behaves

- **Eligibility questions** use follow-up questions (age, state, income and similar details) before answering.
- **Low confidence** leads to 'I'm not sure' and a pointer to the relevant office or helpline, not a guess.
- **Every answer cites its source** and the date it was last updated.
- **Health and legal answers** give information only and recommend an official office or professional for decisions.
- **Personal data** is masked before it is stored or sent to any model.

2.2 Knowledge base preparation (done offline, refreshed regularly)



Scheduled re-crawl keeps schemes and dates current

Official documents are collected, converted to text, split into small sections, tagged with details such as scheme, state, language and date, converted into a searchable form and stored. Stored content is refreshed on a schedule so schemes and dates stay current.

3. Eight-week plan

Week	Focus	Tasks	Deliverable
1	Setup and data	Finalise topic, languages and the two domains. Set up a shared code repository. Try out candidate tools on small samples and choose what works. Start collecting 30 to 50 official documents.	Decisions on tools, languages and domains; document list
2	Knowledge base	Extract text from the documents. Clean and split it into sections. Add details (scheme, state, language, source link). Make the content searchable.	Searchable knowledge base
3	Text-only question answering	Connect search to the language model. Instruct it to answer only from retrieved content, keep answers simple, cite sources and say 'not sure' when unsure. Test with about 20 English questions. Start the report.	Text chatbot with citations
4	Multiple languages	Add language detection and translation. Test in Hindi and the regional language. Fix wrong or awkward answers.	Text chat in 3 languages
5	Voice	Add voice input and spoken replies. Test with real voices and accents. Add a 'please repeat' message for unclear audio.	Voice in and out, 3 languages
6	Web interface	Build the web page with chat, microphone button, language selector and source display. Connect it to the system.	Complete working web app
7	Testing and improvement	Write 30 to 50 test questions per language. Record whether each answer is correct, based on a source and easy to understand. Fix the most common failures. Test off-topic and unsafe questions.	Results table and improved system
8	Report and demo	Write the report and slides. Record a backup demo video. Rehearse the live demo twice. Clean up the code and write the README.	Report, slides, demo video, final code

Protecting the timeline

- **Finish text before voice.** If text works, voice is an extra layer; a good text demo is still possible if time runs short.
- **Add the third language last** and drop it if weeks 4 and 5 run long.
- **Keep the data small.** 30 to 50 good documents beat hundreds of messy ones.
- **Start the report in week 3** and save test results as you go.
- **Hold a 15-minute weekly check-in** and show the mentor something working each week.